Brew Some DEEP LEARNING Using Caffe

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When a human being starts learning in the early stages of life, every vision or simulation that takes place is recorded, compared and analysed almost instantaneously in the mind. It is complex, and we are gifted with a framework that helps us connect, recall and act to events.

In the machine world, intelligence must be taught to otherwise dumb machines. It is now possible to create a framework for them to think like us. Engineers have been working round the clock to perfect this, and as a result, software such as Caffe provide a gateway to deep learning.

Deep learning is a machine learning framework. To imbibe intelligent decision making into machines, deep learning smart algorithms must be devised. In short, it is the class of machine learning algorithm that forms a machine learning framework using which it obtains and processes real-life events.

Techniques include teaching and eventually creating a medium of interaction between man and machine. Machine vision and natural language processing are hot topics today, and any developer can play around with Caffe.

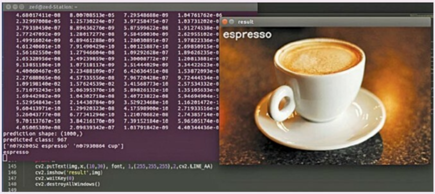

Fig. 1: Caffe installation

It is Caffe and it is brewed

Caffe is a deep learning framework. Its mantle circles around improving visual based image and video processing in optical devices such as cameras. Deep learning techniques can be combined with Caffe framework, and there are limitless possibilities that visual analytics provide. These could be your virtual eyes and brain, among others, for which the credit goes to advancements in sensor technology, high-performance graphic processing and evolving neural networks using advanced algorithms in this field.

How Caffe is brewed. Data in the form of images and videos is what Caffe dines on. Data enters Caffe through data layers that lie at the bottom of nets. In some cases, it is sourced from databases (LevelDB or lightning memory mapped database) directly from memory or from files on the disk in HDF5 format, depending upon the use cases, and Caffe could be used to build models on it.

The models of Caffe work on deep networks that are made up of interconnected layers comprising chunks of data. Caffe has blobs or blob arrays that are used for storing, communicating and manipulating the information. It defines it in a net layer-by-layer model schema. The details of blob describe how information is stored and communicated in and across layers and nets because it is an N dimensional array.

The net follows the collection and connection of layers, which have two key responsibilities: constituting a forward pass for taking inputs and producing an output. The backward pass takes the gradient with respect to output, and computes the gradients with respect to parameters and to inputs, which are, in turn, back-propagated to earlier layers. All you need to do is define the setup, forward and backward passes for the layer, and it is ready for inclusion in a net.

Elements of the brew

- Blob, layers and nets
- Forward and backward passes
- Loss function and loss weights
- Solver snapshotting and resuming
- MatCaffe
- Data for model creation

Vision based use cases

Various uses cases of Caffe include surveillance and vision based alerting using Cloud and computer vision. Vision layers make room for the images from the real world as inputs and analyse colourations and spatial structures. These layers work by applying a particular operation to a region of the input and a corresponding region of the output, treating